

From Plausibility Machines to Narrative Minds

Vision, Attention, and the Architecture of Artificial Cognition

Brook Miller

Center for Humanities and Digital Research, University of Central Florida

July 8, 2026

Outline

- 1 Plausibility Machines and the Bootstrapping Problem
- 2 An Analogy: Human and Computer Vision
- 3 Event Segmentation: A Basis for Experience
- 4 “Attention Is All You Need”
- 5 Coglinks and Uncertainty
- 6 Time, Prediction, and Memory
- 7 Phenomenological Experience and Emotion
- 8 Narrative in Cognition
- 9 What Architecture Is Closer?
- 10 What Training Is Required?
- 11 How Might a Narrative-Processing AI Differ?
- 12 Notes / Parking Lot
- 13 Conclusion

- Since ChatGPT's release (Dec. 2022), a wave of claims that AGI is imminent

- Since ChatGPT's release (Dec. 2022), a wave of claims that AGI is imminent
- Pushback: Chiang and Tufekci — LLMs as **plausibility machines** (Chiang 2024; Tufekci 2026), producing text that reads as coherent without underlying understanding

Long's reframing (Long 2026)

Sneerer: “just predicting the next token” $\Rightarrow$ therefore not conscious

Defender: humans are unreliable and contradictory too $\Rightarrow$ so what's the difference?

- Since ChatGPT's release (Dec. 2022), a wave of claims that AGI is imminent
- Pushback: Chiang and Tufekci — LLMs as **plausibility machines** (Chiang 2024; Tufekci 2026), producing text that reads as coherent without underlying understanding

Long's reframing (Long 2026)

Sneerer: “just predicting the next token” $\Rightarrow$ therefore not conscious

Defender: humans are unreliable and contradictory too $\Rightarrow$ so what's the difference?

The more interesting question

Not *whether* LLMs and humans differ, but precisely *how* — and what that implies.

The Bootstrapping Problem

Long (Long 2026)

LLMs “learned language first and would bootstrap any potential experiences from that, whereas humans can have experiences independently of using language to describe them.”

The Bootstrapping Problem

Long (Long 2026)

LLMs “learned language first and would bootstrap any potential experiences from that, whereas humans can have experiences independently of using language to describe them.”

- It may make sense to ask what it is like to be a bat; asking the same of an LLM is dubious

The Bootstrapping Problem

Long (Long 2026)

LLMs “learned language first and would bootstrap any potential experiences from that, whereas humans can have experiences independently of using language to describe them.”

- It may make sense to ask what it is like to be a bat; asking the same of an LLM is dubious
- **What an LLM is:** procedural circuits — comparable to functions in a CS sense

Claim

If we want AI with a closer semblance to human cognition, a language-first architecture may be the wrong place to start. Look elsewhere.

The Bootstrapping Problem

Long (Long 2026)

LLMs “learned language first and would bootstrap any potential experiences from that, whereas humans can have experiences independently of using language to describe them.”

- It may make sense to ask what it is like to be a bat; asking the same of an LLM is dubious
- **What an LLM is:** procedural circuits — comparable to functions in a CS sense
- **What is (partly) missing:** valence and salience

Claim

If we want AI with a closer semblance to human cognition, a language-first architecture may be the wrong place to start. Look elsewhere.

An Analogy: Human and Computer Vision

- **Edge detection** — low-level feature extraction

An Analogy: Human and Computer Vision

- **Edge detection** — low-level feature extraction
- **Abstraction** — features aggregate into intermediate representations

Why look here

Vision research offers embodied, non-linguistic experience-formation to study — a candidate escape from the bootstrapping problem. Raw input → abstraction → recognition is a useful template for thinking about cognitive architecture generally. Where does narrative fit in this pipeline?

An Analogy: Human and Computer Vision

- **Edge detection** — low-level feature extraction
- **Abstraction** — features aggregate into intermediate representations
- **Object recognition** — abstraction resolves into a labeled category

Why look here

Vision research offers embodied, non-linguistic experience-formation to study — a candidate escape from the bootstrapping problem. Raw input → abstraction → recognition is a useful template for thinking about cognitive architecture generally. Where does narrative fit in this pipeline?

- Waymos endlessly circling roundabouts and neighborhoods — fair grounds for skepticism

Oh No, Waymo

- Waymos endlessly circling roundabouts and neighborhoods — fair grounds for skepticism
- But moving, embodied AI does not *ipso facto* share the LLM's bootstrapping problem

Slow feature analysis (Song and Zhao 2022)

A baseline for experiential processing — it builds temporal parsing into experiencing.

Oh No, Waymo

- Waymos endlessly circling roundabouts and neighborhoods — fair grounds for skepticism
- But moving, embodied AI does not *ipso facto* share the LLM's bootstrapping problem
- Regulating a body moving through space is an ongoing, dynamic problem: recognizing other agents and dynamic forces in the world

Slow feature analysis (Song and Zhao 2022)

A baseline for experiential processing — it builds temporal parsing into experiencing.

Event Segmentation

What an event is, minimally

“a segment of time at a given location that is perceived by an observer to have a beginning and an end” (Miller 2023, p. 128)

Event Segmentation

What an event is, minimally

“a segment of time at a given location that is perceived by an observer to have a beginning and an end” (Miller 2023, p. 128)

Event Segmentation Theory (EST)

“One central proposal of the EST model is that event segmentation is a spontaneous outcome of ongoing perception and arises at points where perceptual predictions begin to fail or change. . . The period of epoch between two such consecutive changes current mental models is an event.” (Miller 2023, p. 135)

Prediction, Revisited

- Sneerer/defender critique centers on LLMs' prediction as reductive: next token, given previous tokens

Prediction, Revisited

- Sneerer/defender critique centers on LLMs' prediction as reductive: next token, given previous tokens
- But perceptual, event-driving prediction is different in kind:

“As an event proceeds, predictions are relatively accurate. However, as it ends, prediction error increases, causing a reset of the event model. When such a reset occurs, an event boundary is set.” (Miller 2023, p. 135)

Prediction, Revisited

- Sneerer/defender critique centers on LLMs' prediction as reductive: next token, given previous tokens
- But perceptual, event-driving prediction is different in kind:

“As an event proceeds, predictions are relatively accurate. However, as it ends, prediction error increases, causing a reset of the event model. When such a reset occurs, an event boundary is set.” (Miller 2023, p. 135)

This requires online processing of ongoing experience

Two research threads follow: **slow feature analysis** and **uncertainty processing**.

“Attention Is All You Need” (Vaswani et al. 2017)

- What attention consists of, for an LLM
- Stacked transformers and attention
- Loss minimization and static associative valuation

Attention, Contrasted: Weights vs. Awareness

- Transformer attention: a computed weighting over tokens in a context window — not modeled as an object of the system's own awareness

Attention, Contrasted: Weights vs. Awareness

- Transformer attention: a computed weighting over tokens in a context window — not modeled as an object of the system's own awareness
- Webb and Graziano's **Attention Schema Theory** (Webb and Graziano 2015): awareness is *an internal model of attention*

“A creature with an attention schema...”

“...should be a creature that concludes it is aware” (Webb and Graziano 2015)

Attention, Contrasted: Weights vs. Awareness

- Transformer attention: a computed weighting over tokens in a context window — not modeled as an object of the system's own awareness
- Webb and Graziano's **Attention Schema Theory** (Webb and Graziano 2015): awareness is *an internal model of attention*

“A creature with an attention schema...”

“...should be a creature that concludes it is aware” (Webb and Graziano 2015)

Open possibility

Could an explicit, internal model *of the system's own attention* — not just attention weights themselves — point toward AI architectures with a closer semblance of awareness? (cf. Section “What Architecture Is Closer?”)

Second research thread, promised above

Uncertainty processing in hierarchical decision-making (Wang, Lynch, and Halassa 2025)

Second research thread, promised above

Uncertainty processing in hierarchical decision-making (Wang, Lynch, and Halassa 2025)

- Circuits imitated
- Predefined decision choices

Time, Prediction, and Memory

- Predictive processing vs. uncertainty
- Episodic and semantic memory

Phenomenological Experience and Emotion

Open question

What would have to be true of a system for phenomenological experience and emotion — not merely their outward signatures — to be present?

Narrative in Cognition (Miller 2024)

- As **active inference**

Narrative in Cognition (Miller 2024)

- As **active inference**
- As an **abstraction scaffold**

What Architecture Is Closer?

- Self-salience models
- Less terrifying possibilities that simulate emotion

What Training Is Required?

- Direction 1: [To be developed]
- Direction 2: [To be developed]

How Might a Narrative-Processing AI Differ?

- 1 Not procedural?

How Might a Narrative-Processing AI Differ?

- ① Not procedural?
- ② Predictive processing (with respect to dynamic inputs)?

Thus

A closer semblance of self.

How Might a Narrative-Processing AI Differ?

- ① Not procedural?
- ② Predictive processing (with respect to dynamic inputs)?
- ③ Nested experiential hierarchies — including reflection on what experiences reveal about self-function?

Thus

A closer semblance of self.

- **Waymo, continued** —

valuative processing and attention, beyond slow feature analysis

Slow feature analysis combines with valuative processing (salience and valence) that is itself dynamic and interactive. In terms of attention, a stabilization in valuation can be a catalyst of shifting attention, which marks temporal experience with subjective meaning.

Open questions: Can an experiencing AI have a semblance of these processes? How might it relate to the question of whether the AI has subjective experience?

- **Hebbian learning vs. gradient descent** —

Hebbian as local; gradient descent as global

Thank you!

Questions?

References I

-  Chiang, Ted (2024). *PLACEHOLDER – Ted Chiang on LLMs as “plausibility machines” (verify exact source)*. Placeholder citation — fill in exact title, publication, and date.
-  Long, Robert (Apr. 2026). *Language models are different from humans, and that’s okay. Experience Machines* (Substack). Published April 1, 2026; light edit April 2, 2026.
-  Miller, Brook (2023). “Events and Weak Narrativity”. In: *Narrativity in Cognition*. Chapter 6; starts p. 119. Quoted material below drawn from Sect. 6.2 “Attention and Events,” pp. 124–130. Springer Nature Switzerland. DOI: 10.1007/978-3-031-40349-1_6.
-  — (2024). *Narrativity in cognition*. Springer.
-  Song, Pengyu and Chunhui Zhao (2022). “Slow down to go better: A survey on slow feature analysis”. In: *IEEE Transactions on Neural Networks and Learning Systems* 35.3, pp. 3416–3436.
-  Tufekci, Zeynep (2026). “The Unstoppable Force of A.I. Hype Is Meeting One Immovable Fact”. In: *The New York Times*. Opinion, Guest Essay; published June 30, 2026.

References II

-  Vaswani, Ashish et al. (2017). “Attention is all you need”. In: *Advances in Neural Information Processing Systems* 30.
-  Wang, Mien Brabeeba, Nancy Lynch, and Michael M Halassa (2025). “The neural basis for uncertainty processing in hierarchical decision making”. In: *Nature Communications* 16.1, p. 9096.
-  Webb, Taylor W. and Michael S. A. Graziano (2015). “The attention schema theory: A mechanistic account of subjective awareness”. In: *Frontiers in Psychology* 6. Article 500. DOI: 10.3389/fpsyg.2015.00500.